

to explain the equity premium, however, it is exactly the opposite. In Bansal and Yaron's world, almost all dividend yield variation comes from cash flows, and there is essentially no discount rate channel.¹⁹ The risk premiums are still predictable according to equation (8.4), but they are driven only by predictable cash flows.

I have a stake in this literature and, unlike such "true believers" in discount rate predictability as Cochrane, my research documents the existence of cash flow predictability.²⁰ In Ang and Liu (2007), I show that since both dividend yields and equity return volatility vary over time (see also section 6 below), it *must* be the case that risk premiums also exhibit predictable components. Overall, my opinion is that the truth lies somewhere in between the Cochrane and the Bansal–Yaron extremes: there is both cash flow and discount rate predictability, but it is hard to detect both (see below).

What an investor should take away is that dividend yields vary over time, and, as a result, expected returns and cash flows are predictable. In the discount rate story, low prices mean high returns in the future and times of low prices are good times to buy equities. If we can forecast high future cash flows according to the cash flow story, we want to hold equities. A smart investor might be able to use these facts to her advantage.

5.2. THEORY SAYS TIME-VARYING EQUITY RISK PREMIUMS ARE HARD TO ESTIMATE

Theory says equity risk premiums are predictable, but theory also says that the amount of predictability is small. Thus, it is hard to statistically detect predictability in data and even harder to trade on it.

Predictability Regressions

A common statistical model to capture predictability is a *predictability regression*:

$$r_{t+1} = c + b \cdot X_t + \varepsilon_{t+1}, \quad (8.6)$$

where r_{t+1} is the market excess return and X_t is a set of predictive variables. The predictors range from the economically intuitive factors, like macro factors, to more esoteric variables, like sentiment-based factors based on Twitter or Facebook.²¹ In regression (8.6), the strength of predictability is measured by the statistical significance of the predictive coefficient, b . Econometricians like to see high t -statistics, or low p -values, with a standard cut-off of a p -value of 5% called "statistically significant." If the returns and the predictive instruments have been standardized, that is, set to zero mean and a variance of one, then b represents

¹⁹ There is a discount rate channel in Bansal and Yaron because there is stochastic volatility and the risk-free rate is time varying. But the variation in these components is small.

²⁰ See Ang and Bekaert (2007), Ang (2012b), and Ang and Zhang (2012).

²¹ See Bollen, Mao, and Zeng (2011) and Karabulut (2011), respectively.